NeurIPS 2026

MSR-3D: Multi-Mode Semantic Representation for Open-Vocabulary 3D Scene Understanding

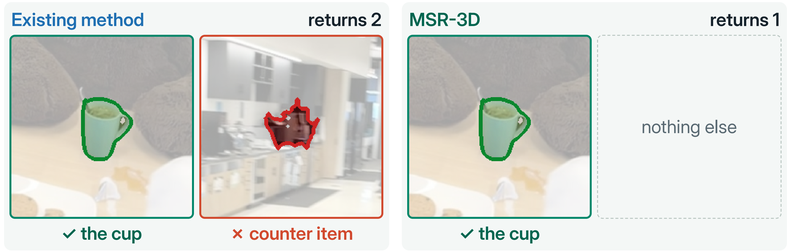
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Code · weights · data

KEY IDEA Don't average an object's views into one feature. Keep one prototype per appearance mode, and learn to match queries against all of them.

1 One feature lets wrong objects in

Query **coffee** in a scene unseen in training (teatime). Existing methods compress all views of an object into **one feature** and match the query against it.

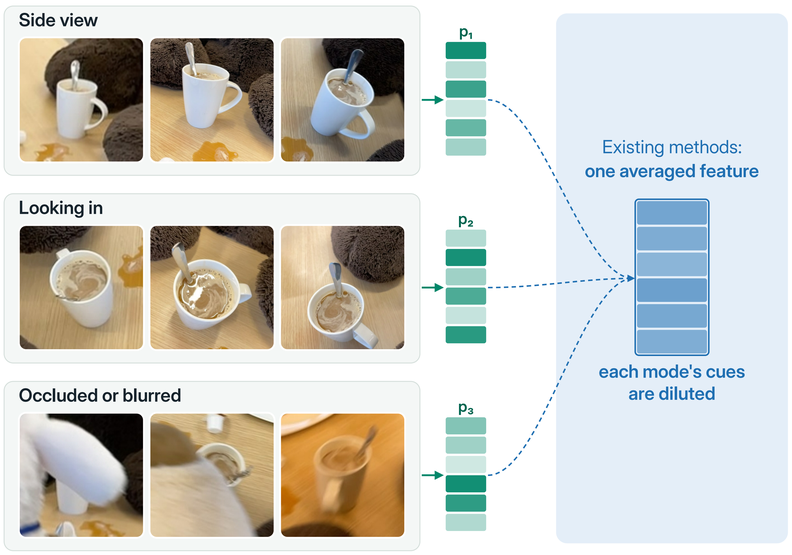


Existing method shown: OpenSplat3D, the strongest baseline we compared. Each tile: the same frame (#171), zoomed on one returned object.

One similarity score lets related but wrong objects through.

2 Idea: one prototype per appearance mode

Across its 127 views the cup looks different. Averaging their features mixes these modes together.



Keep the modes apart instead of averaging them.

3 How MSR-3D works

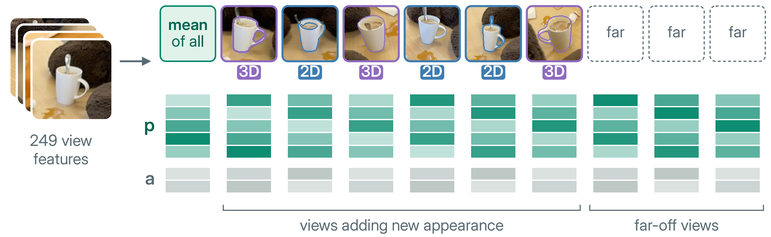
a Observe each view through two masks



Each view is encoded under the 2D SAM mask and the rendered 3D object.

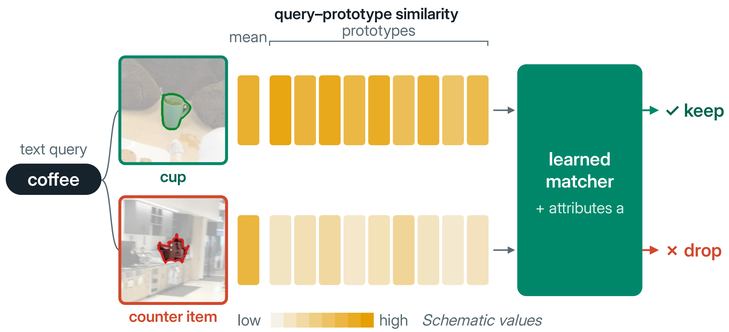
They fail differently: under occlusion SAM follows the toy, the render keeps the cup (bottom row).

b Compress the views into prototypes



This cup: 249 view features → **10 prototypes**. Each keeps a feature **p** and attributes **a** (e.g. support, 2D/3D source).

c Learn to match a query



A small learned matcher reads the query's similarity to every prototype together with the attributes, and keeps or drops each object.

Trained once on 18 queries of one scene; reused unchanged everywhere else.

4 Results

mIoU gain over the strongest baseline (points)

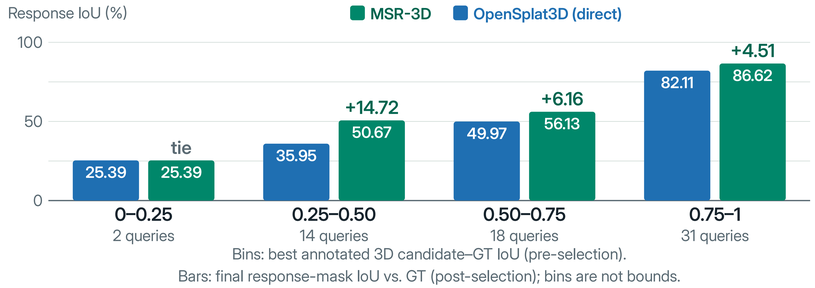
+6.69	+3.46	+1.57
LERF67 (4 scenes)	LERF45 (unseen query strings)	3DOVS61 (fixed parameters)

mIoU / mF1 (%)	LERF67	LERF45	3DOVS61
OpenSplat3D	58.01 / 61.07	59.69 / 65.68	79.41 / 80.52
Segment-then-Splat	50.74 / 58.93	56.29 / 66.19	76.12 / 81.11
MSR-3D	64.70 / 72.63	63.15 / 72.83	80.98 / 83.02

LERF67: 4 scenes incl. supervised Waldo. **LERF45:** 3 other scenes, query strings not among the 18 supervised ones. **3DOVS61:** 10 scenes, 61 queries. Shared inputs; equal scene weights; Segment-then-Splat selects one object.

5 Bigger gains on imperfect candidates

LERF-OVS queries grouped by how good the best upstream 3D candidate is, before any matching.



Not shown: 2 queries with no positive candidate (1.86 vs 0.92). 3DOVS: 47 of 61 queries fall in the top bin (89.52 vs 87.95).

MSR-3D makes better use of imperfect candidates.

6 Many instances: "knife"



Waldo kitchen (supervised scene), 6 knives; green found, orange missed. OpenSplat3D's 4 wrong picks lie outside this view; one knife is not visible here.